

Section 4.6 Exercise Answers

1. $v = -1$

2. $x = -\frac{1}{3}$

3. $x = \frac{4}{3}$

4. $n = -1$

5. $x = -\frac{4}{3}$

6. $b = \frac{6}{5}$

7. $n = -\frac{1}{6}$

8. $x = 10$

9. $x = \frac{\ln(\frac{13}{2})}{6}$

10. No solution

11. $a = \frac{\log(\frac{29}{2})}{9}$

12. $p = \log\left(\frac{17}{8}\right) - 7$

13. No solution

14. $k = -\frac{\ln(38)}{3}$

15. $x = \frac{\ln(\frac{54}{5}) + 8}{9}$

16. $x = \frac{\ln(\frac{38}{3}) - 8}{9}$

17. $x = \frac{-\ln(2) - \ln(5)}{\ln(2) - 2\ln(5)}$

18. $x = \ln(12)$

19. No solution

20. $x = \frac{\ln(\frac{3}{5}) - 3}{8}$

21. $x = \frac{\ln(15) - 3}{3}$

22. No solution

23. $x = \frac{-2\ln(7) - \ln(3)}{2\ln(3) - \ln(7)}$

24. $x = \ln(3)$

25. No solution

26. $n = 49$

27. $x = \frac{1}{81}$

28. $k = \frac{1}{36}$

29. $n = 12$

30. $x = \frac{9-e}{8}$

31. $x = -10$

32. $n = 1$

33. $x = \frac{3}{73}$

34. No solution

35. $m = \frac{3}{2}$

36. No solution

37. $n = 9$

38. $x = \pm \frac{10}{3}$

39. $x = \frac{12}{11}$

40. $x = 10$

41. $x = \frac{2}{7}$

42. $x = 0$

43. $x = \frac{2}{19}$

44. $x = \frac{3}{4}$

45. $x = 154$

46. Approximately 5 years

47. $14,163 \text{ ft}$